

Cardiology Handover — Saroj Agarwal

Prepared for: Dr. Nilesh Purushottam, Cardiologist, Surat **Date:** 20 July 2026 **Prepared by:** Family (son), from the primary hospital records listed in §10 **Referral path:** Dr. Talha Meeran (Consultant Cardiologist, Sir H. N. Reliance Foundation Hospital, Mumbai — her treating cardiologist through the Feb–Apr 2026 admission and afterwards) advised a **local Surat review by Dr. Nilesh Purushottam**.

Every figure below is quoted from a source document; the file for each is named in §10. Where the records are silent or contradict each other, that is stated explicitly rather than smoothed over.

PAGE 1 — AT A GLANCE

Saroj Agarwal, F/63 (DOB 27-Aug-1962). Metastatic Ca endometrium, on chemo-immunotherapy, currently in partial response (PET-CT 22 Jun, –47.1 %).

The cardiac story in six lines:

1. **Severe pulmonary arterial hypertension** since Feb 2026, judged **Group 3** (secondary to lung/pleural disease). RVSP **83 → 72 → 91 (at PE) → 70 mmHg** (06 Jul). Never right-heart-catheterised.
2. **Ventricular arrhythmia, Mar 2026** — multifocal PVCs, bigeminy, episodes of **polymorphic PVCs/VT**. Acute treatment lignocaine infusion; then maintained on **amiodarone + mexiletine ± bisoprolol**.
3. **Acute submassive saddle pulmonary embolism, 24–25 Mar 2026 → mechanical thrombectomy** → complicated by shock, haemoptysis, platelets to 51, Hb to 7.3. Anticoagulated since. **No residual pulmonary thrombus** on PET-CT 22 Jun.
4. **LVEF 55–60 % (Feb–Mar) → ~50 % (Jul)**, new mild concentric LVH, grade I diastolic dysfunction. **RV function has recovered** (was FAC 20 % in March; normal now).
5. **QTc 481–529 ms across five ECGs this month**, on amiodarone + mexiletine. **This is the reason for the visit.**
6. **72-hour Holter 10–13 Jul was clean** — sinus, **0 VT runs**, 0 pauses, 0 AF, 0.1 % ectopic burden, and **her symptomatic episode occurred in normal sinus rhythm**.

Current cardiac drugs — confirmed from the **Maitreya discharge summary of 17 Jul 2026**: Apixaban (Apixalife) 5 mg **BD** · Amiodarone (Cordarone) 100 mg **BD** · Mexiletine (Mexohar) 150 mg **BD** · Spironolactone (Aldactone) 50 mg **OD** · **Torsemide (Dytor) 20 mg BD** · Tadalafil 20 mg **HS**. **No beta-blocker**. Also on a **tapering prednisolone course** (40→20→10→5 mg) and **metoclopramide SOS**.

The proposal under review: Dr. Narendra Tanwar (Maitreya, Surat) suggests **stopping mexiletine as pro-arrhythmic and adding a beta-blocker**. Dr. Talha Meeran (RFH Mumbai, her cardiologist throughout) referred her to you for this decision.

Six things from the record that bear on that decision:

- **A. Amiodarone — not mexiletine — is the QT-prolonging drug.** Mexiletine (class IB) tends to *shorten* QT; it is used therapeutically in long-QT type 3.
- **B. Much of the “prolonged QTc” is a Bazett artefact.** All five July ECGs are Bazett-corrected at **HR 84–94**, where Bazett over-corrects. Raw QT is only **377–440 ms**. On **Fridericia** the same traces read **492 → 432 → 490 → 462 → 450 ms**.
- **C. ★ The doses were already cut ~13 Jul — and the QT is responding.** Amiodarone was halved (200→100 mg **BD**) and mexiletine cut from TDS to **BD** by the Maitreya team **with Dr. Meeran’s approval**. Since then: **QTc 529 → 491 → 481 ms** (Fridericia 490 → 462 → **450**). An intervention is already in flight and working.
- **D. A beta-blocker was already tried — and dropped.** Concor (bisoprolol) ran **06–20 Mar 2026**, then **vanished from the 08 Apr discharge list, with no reason documented**. She has recurrent bronchospasm and a pressure-loaded RV.
- **E. Two current drugs are actively depleting her electrolytes.** **Torsemide 20 mg BD** (wastes K^+ and Mg^{2+}) plus **tapering prednisolone** (wastes K^+), while **K^+ is 3.57–3.89** against her own team’s **$K^+ > 4$ / $Mg^{2+} > 2$** target — and **no magnesium level has been measured since 08 Mar 2026**. **Perinorm (metoclopramide) SOS is itself QT-prolonging**.
- **F. Her cardiac biomarkers are reassuring.** Troponin has **never** been raised — 3.2 (Mar), **11.90 even during the saddle PE** (ref ≤ 15.6). NT-proBNP down from 4089 → **151** (05 Jul).

Never done: right heart catheterisation · coronary angiogram (considered 08 Mar, **deferred**) · thyroid/liver monitoring in 4.5 months of amiodarone.

Competing explanation for her current symptoms: the **05–08 Jul admission was for fever + hypoxia (SpO_2 84 % on air) — viral LRTI with volume overload**, not arrhythmia; “Prolonged QT interval” was a *listed diagnosis*, not the reason for admission. HRCT 05 Jul shows **pleural effusions re-accumulated and increased**. She left 08 Jul **off oxygen** but was discharged 17 Jul **back on 1 L/min O_2** . Clean Holter + normal troponin + near-normal NT-proBNP + recovered RV all point away from the heart.

1. Patient identifiers

Name	Saroj Agarwal (also spelt Agrawal on Surat reports)
Sex / Age	Female, 63 years
DOB	27 August 1962
ID — RFH Mumbai	0010663186
ID — Maitreya Surat	UHID 59568
Oncological diagnosis	Carcinoma of uterus / endometrium (ICD-10 C54) — operated ≈ 2023, metastatic recurrence Feb 2026

2. Why this consultation

Dr. Narendra Tanwar (DM Cardiology, Maitreya Multispeciality Hospital, Surat) has noted that her **QT interval remains prolonged**.

- **Current antiarrhythmics (confirmed, 17 Jul discharge summary):** Mexohar (mexiletine) 150 mg **BD** + Cordarone (amiodarone) 100 mg **BD**.
- **Already done ~13 Jul:** both were **reduced** (amiodarone halved from 200 mg BD; mexiletine cut from TDS to BD) by the Maitreya team **with Dr. Meeran's approval**, in response to the QT. The QTc has fallen on each ECG since.
- **Local suggestion now:** stop mexiletine on the grounds that it is pro-arrhythmic, and add a **beta-blocker**.
- **Second opinion sought** from Dr. Meeran, who has all the historical context; he has referred her locally to you.

3. Active cardiac problem list

1. **Severe pulmonary arterial hypertension** — present from day 1 of the Feb admission, attributed by the RFH team to underlying lung pathology (**Group 3**). RVSP ~70 mmHg on the most recent echo (06 Jul 2026). Never right-heart-catheterised.
2. **Ventricular arrhythmia, currently quiescent** — frequent multifocal PVCs, ventricular bigeminy, and episodes of **polymorphic PVCs / VT** in Mar 2026, treated acutely with lignocaine (Xylocard) then maintained on amiodarone + mexiletine ± bisoprolol. A 72-hour Holter in Jul 2026 was **clean**.
3. **Prolonged QTc, 481–529 ms across five ECGs this month**, on amiodarone + mexiletine — **improving since the doses were cut ~13 Jul**. Formally listed as a discharge diagnosis (R94.31).
4. **Prior acute submassive saddle pulmonary embolism (24–25 Mar 2026)** — mechanical thrombectomy, then anticoagulated. **No residual pulmonary arterial thrombus** on the 22 Jun 2026 PET-CT.
5. **Mild LV systolic decline** — LVEF 55–60 % (Feb–Mar) → **~50 %** (Jul), with new mild concentric LVH and grade I diastolic dysfunction.
6. **Re-accumulating bilateral pleural effusions** (HRCT 05 Jul) — the likely driver of her current breathlessness.
7. **Ischaemia never formally excluded** — coronary angiography was considered on 08 Mar 2026 and **deferred**; no angiogram has ever been done. See §8.

4. Chronological narrative

4.1 Before the Mumbai admission

- Carcinoma of uterus, operated ~3 years prior (≈ 2023).
- Feb 2026: recurrence with **left pleural effusion and lung collapse**; intercostal drain in situ from **12 Feb 2026** at an outside hospital. Pleural fluid cytology (14 Feb) — **atypical cells suggestive of carcinoma, large-cell morphology**. CA-125 635.50 (14 Feb). NT-proBNP 1738 high and CPK-MB 34.67 high on 11 Feb, with Troponin-I negative.

4.2 Sir H. N. Reliance Foundation Hospital, Mumbai — 20 Feb to 8 Apr 2026 (48 days)

Transferred in with breathlessness, cough and bilateral pedal oedema. Attending: Dr. Darshit Shah (Medical Oncology). Cardiology: Dr. Talha Meeran.

20 Feb — severe PAH found on day 1. Echo: **RVSP 83 mmHg**, LVEF 55 %, mildly dilated RV with fair systolic function (TAPSE 16 mm), grade I diastolic dysfunction, mild TR. Chemotherapy started same day (Paclitaxel + Carboplatin, C1D1). NT-proBNP 1761; INR 2.47 / PT 27.4 s at baseline, before any anticoagulant.

24–26 Feb — airway crisis. Stridor; nasopharyngolaryngoscopy showed **bilateral vocal cords in paramedian position with restricted mobility** and minimal glottic chink. Desaturation episode with **irregular pulse** noted (25 Feb, MET call). On 26 Feb she underwent **left vocal cord lateralisation** under GA; desaturated intraoperatively, required intubation and ICU ventilation. Pre-op echo: RVSP 72 mmHg, LVEF 60 %. The PAH was formally attributed to **underlying lung pathology (Group III)**.

01 Mar — echo on ICU: LVEF 60 %, mildly dilated RV with normal function, mild TR, **PASP ~46 mmHg**. Telemetry: **sinus rhythm with intermittent ventricular bigeminy**. > ⚠️ This ~46 mmHg figure appears only inside the discharge-summary narrative for **01 March**; there is no standalone echo report for it in the records. (An earlier internal note in this archive mis-dated it to “early April” — that is wrong.)

06 Mar — the arrhythmia problem begins. Cardiology review noted **frequent PVCs and NSVT**; antiarrhythmic therapy started — **Cordarone (amiodarone) + Concor (bisoprolol)**.

08 Mar — the worst arrhythmic day. Episodes of **polymorphic PVCs and VTach with palpitations**, managed with a **Xylocard (lignocaine) infusion** plus antiarrhythmics. **Coronary angiography was considered but deferred after cardiology discussion**. Same day: hs-Troponin I **3.2 pg/mL (normal, ref ≤15.6)**; Magnesium 2.21 mg/dL (normal).

09–13 Mar — acute bronchospasm 09 Mar (likely microaspiration), treated with steroids. Arrhythmias settled. Regimen through this period: **Concor, Cordarone, Mexiletine**, Lasix, Aldactone, Tadalafil. IV potassium and magnesium correction given, with an explicit target of **K⁺ > 4 and Mg²⁺ > 2** — i.e. the team was already managing her electrolytes with the ventricular irritability in mind.

14–17 Mar — 72-hour Holter (RFH). Ordered by Dr. Meeran. Result: sinus rhythm, avg 75 bpm (52–106). **1811 PVCs (<1 %)**, 1809 isolated, 1 couplet, **60 episodes of ventricular bigeminy, 0 VT runs, 0 pauses, 0 AF, 0 AV block**. Interpretation: “*Underlying normal sinus rhythm with occasional multifocal ventricular ectopies along with brief runs of ventricular bigeminy. No NSVT or VTACH.*” On 17 Mar, ECG **QTc 435 ms** — i.e. her QTc was *normal* at that point on the same drug class.

20 Mar — MET call for chest pain; haemodynamically stable, managed symptomatically. Cardiac drugs continued (Concor, Cordarone, Tadalafil, Mexiletine, Lasix, Aldactone).

23–25 Mar — the pulmonary embolism. Acute breathlessness with **hypoxia, SpO₂ 77–78 % on room air** (23 Mar). Clexane started 24 Mar. - **CTPA (24 Mar):** “*Acute submassive pulmonary embolism with saddle thrombus involving the main pulmonary trunk into the right and left main pulmonary artery, lobar and segmental branches, associated with RV strain and dilated pulmonary arteries.*” Pulmonary trunk 35 mm. Completely occluding thrombus in the proximal left upper lobar segmental branch; partially obstructive defects throughout the right lower, middle and upper lobe branches. Moderate bilateral pleural effusions. - **Echo (24 Mar, Dr. Maulik Parekh, Interventional Cardiology):** large saddle thrombus at the RPA/LPA bifurcation; **RVSP 91 mmHg**; moderate-to-severe TR; **dilated RA (40 mm) and RV (50×45×77 mm) with RV systolic dysfunction, FAC 20 %**; LVEF 60 %. - **NT-proBNP 4089 pg/mL (24 Mar)** — her peak. - **hs-Troponin I only 11.90 pg/mL (25 Mar) — still within normal limits (≤15.6)** despite a saddle PE with RV dysfunction. Worth noting: she has *never* had a raised troponin in these records. - Troponin was checked specifically to decide on **ICU-based thrombolysis**.

~25–26 Mar — mechanical thrombectomy. > ⚠️ **Documentation gap.** The procedure itself is confirmed three times — the discharge summary records “*Post pulmonary thrombectomy status with high PESI risk*” (04 Apr), a blood culture of 31 Mar is labelled “*Clinical Details: Post thrombectomy*” (site: left IJV), and the Maitreya echo of 06 Jul records “*S/P Mechanical Thrombectomy (03/2026)*”. **But there is no procedure note, no exact date and no named operator anywhere in the records.** The family recall a “Dr. Rajesh”; that name does not appear in any document. The only interventional cardiologist named anywhere is **Dr. Maulik Parekh**, who reported the 24 Mar echo. *Please request the procedure note from RFH if the technical detail matters.*

26 Mar – 3 Apr — post-thrombectomy complications. Intubated with **persistent circulatory shock requiring vasopressors** (noradrenaline + vasopressin); **ET-tube bleeding and haemoptysis**; **Hb fell 12.9 → 7.3 g/dL** and **platelets fell 109 → 51 ×10⁹/L**; anticoagulation withheld, then cautiously restarted 31 Mar as platelets recovered. Transfused. Extubated 28 Mar. New right middle/lower lobe consolidation with fever 31 Mar — broad-spectrum antibiotics (Zavicefta, Aztreonam); blood cultures **no growth**. Acute kidney injury: creatinine peaked **1.74 mg/dL (27 Mar)**, recovered to 0.71 by 29 Mar. Vasopressors off by 02 Apr.

06 Apr — chemotherapy C3D1 given. 07 Apr — desaturation to 84–85 % after a barium swallow study; discharge deferred. **08 Apr — discharged, haemodynamically stable**, on 1–2 L/min oxygen.

4.3 Interval, April–June 2026 — recovery

- **13 Apr —** daycare chemotherapy (C3D8 Paclitaxel + Carboplatin + C2 Dostarlimab). Cardiac regimen unchanged (§6).
- **01 May —** right-leg swelling investigated. **Right lower limb arterial & venous Doppler: NO DVT.** Veins compressible, normal phasic flow. Swelling attributed to **moderate-to-severe subcutaneous oedema**, with subtle superficial petechial/haemorrhagic change in the right thigh consistent with her anticoagulation. Incidental **mild atheromatous plaque and intimo-medial thickening** in the leg arteries.
- **NT-proBNP normalised:** 4089 (24 Mar) → **97.2 pg/mL (15 Jun)**.
- **22 Jun — follow-up whole-body FDG PET-CT. RECIST 1.1 Partial Response, –47.1 %.** Marked regression of cervical, mediastinal, abdominal and inguinal nodal disease and of peritoneal/omental deposits; interval reduction in bilateral malignant pleural disease. Critically for you: “**No residual pulmonary arterial thrombus**” on the CT component. Last chemo dose 15 Jun 2026.

4.4 July 2026 — Maitreya Hospital, Surat (two separate admissions)

Admission 1 — 05 → 08 Jul 2026 (IPD I/0726/907), under Dr. Sanjay Agrawal, Physician

Correction to an earlier assumption: this admission is often described as being “for chest pain / syncope”. The discharge summary shows it was not. It was an **emergency admission with fever and hypoxia**.

Presentation (verbatim): high-grade fever with cough and whitish expectoration, breathlessness, nausea and vomiting, mild abdominal pain, **bilateral pedal oedema +**, generalised weakness. On arrival: conscious, oriented, **hypoxic and febrile — Temp 101 °F, SpO₂ 84 % on room air**, BP 110/70, HR 100/min. Chest: bilateral rhonchi with left lower zone signs. **Bilateral pedal oedema +++**.

Formal discharge diagnoses (ICD-10): Viral LRTI (J22) · B/L pleural effusion (J90) · Endometrial adenocarcinoma (C54.1) with lung metastasis (C78.00) · Pulmonary embolism (I26) · Pulmonary hypertension (I27.0) · **Prolonged QT interval (R94.31)**.

Course: admitted to **ICU** on oxygen for hypoxia. Bedside 2D echo showed good LV function; **diagnosed with volume overload** and treated with **antibiotics + a Lasix infusion**, nebulisation and supportive care. Improved steadily; shifted to ward 06 Jul; **oxygen tapered off entirely by 07 Jul**, saturating off oxygen. Discharged 08 Jul, “Improved”.

Also from this admission: she was given **IV potassium chloride x2 and IV magnesium sulphate x1 on 06 Jul** (per the pharmacy invoices) — the only magnesium administration on record since March.

Her history as recorded here: endometrial adenocarcinoma **operated 2022**; lung metastasis; **6 cycles of chemotherapy and 6 cycles of immunotherapy** completed; past history of pulmonary embolism.

- **05 Jul — HRCT chest (Navkar):** moderate bilateral pleural effusions, left > right, **INCREASED** in volume vs the 22 Jun PET-CT (right 150–180 cc, left 180–200 cc). **No acute infectious pneumonia. No suspicious lung nodule.** Chronic subsegmental atelectasis LLL + lingula. **Pulmonary artery mildly prominent at 3.4 cm** — consistent with PAH. No pericardial effusion; cardiac size normal.
- **05 Jul — admission bloods:** Hb **11.4** (best since Feb), WBC 7270, platelets 187 000. **Creatinine 1.08** (top of range). Na 136, K **3.57**. **CRP 9.0 mg/L (high)**, ESR 18 (high), N/L ratio 4.38 (high), lymphopenia. **NT-proBNP 151.4 pg/mL** — mildly above the 125 cut-off, far below her Feb/Mar peaks.
- **06 Jul — 2D Echo (Dr. Narendra Tanwar):** Mild concentric LVH; no RWMA at rest; jerky IVS; **LVEF ~50 %**; grade I diastolic dysfunction; mild MR; **mild-to-moderate TR; SEVERE PAH, RVSP/PASP 70 mmHg**; mildly dilated RA and RV with **normal RV function**; IVC 12 mm, good collapse; no clot, vegetation or pericardial effusion. Report carries the note: “Normal Echo does not rule out coronary artery disease.”
- **07 Jul — creatinine 1.20 mg/dL** (above range), Na 135, K 3.89.
- **06 Jul — iron studies:** Iron 55.7, TIBC 310, transferrin sat 17.96 %, ferritin 60.7 — **no iron deficiency**. Urine routine normal.

10–13 Jul — 72-hour Holter (Predicardio, prescribed by Dr. Talha Meeran; indications **syncope, chest pain**). **This is the single most reassuring investigation in the file:** - Predominant rhythm **sinus**. 382 391 beats. HR min 78 / avg 88 / max 115. Longest RR 1.07 s. - Ventricular ectopics **512 total (0.1 % burden)**, 502 isolated, 5 couplets. **0 VT runs**. - **No supraventricular ectopics. No atrial fibrillation. No SVT. No AV block. No pauses >2.5 s**. - **One patient-triggered event, 11 Jul 09:00** — “shortness of breath, weakness/fatigue, chest discomfort/pain” — **the rhythm during and around that episode was normal sinus rhythm**. - Compare with March: PVC burden fell from 1811 to 512, and bigeminy runs largely resolved.

Admission 2 — 13 → 17 Jul 2026 (IPD I/0726/1017, DOA 13 Jul 2:24 pm)

Chemotherapy appears to have been given during this admission (**Inj Frastim/filgrastim 300 mcg dated 14-07-2026** on the discharge prescription — G-CSF is given ~24 h post-chemo). **Discharged 17 Jul on 1 L/min oxygen, SpO₂ 93 %, BP 130/80, PR 83** — i.e. **she has become oxygen-dependent again** since leaving the previous admission off oxygen on 08 Jul.

- **ECG 13 Jul 14:52:** sinus 86 bpm, PR 166, QRS 86, **QT/QTc 410/491 ms**. Large PtfV1, poor R-wave progression V2, inverted T V3, **RVH, prolonged QT**, low voltage in chest leads, indeterminate axis. Confirmed by Dr. Narendra Tanwar.
- **ECG 17 Jul 10:58:** sinus 87 bpm, PR 143, QRS 94, **QT/QTcBz 398/481 ms**. Large PtfV1, **prolonged QT**, low voltage limb leads, indeterminate axis. > ⚠ The 17 Jul trace carries **no patient name** and prints the machine default “Male”; it is attributed to her on the family’s account alone. Please confirm against the hospital’s copy.

5. Objective data — trends at a glance

5.1 Echocardiography

Date	RVSP / PASP	LVEF	RV	TR	Reporter
20 Feb 2026	83 mmHg	55 %	mildly dilated, fair function, TAPSE 16	Grade I	Dr. Talha Meeran
26 Feb 2026	72 mmHg	60 %	mildly dilated, normal function, TAPSE 21	Mild	Dr. Talha Meeran
01 Mar 2026 *	~46 mmHg	60 %	mildly dilated, normal function	Mild	narrative only — no report
24 Mar 2026 (PE)	91 mmHg	60 %	dilated 50×45×77, RV dysfunction, FAC 20 %	Mod-severe	Dr. Maulik Parekh
06 Jul 2026	70 mmHg	~50 %	mildly dilated, normal function	Mild-mod	Dr. Narendra Tanwar

* From the discharge-summary narrative for 01 Mar; no standalone echo report exists.

Read: PAH is better than the March PE peak but **has not returned to the ~46 mmHg nadir** — it has re-risen in step with the re-accumulating pleural effusions. RV function has *recovered*. LVEF has drifted 55–60 % → 50 % (different labs/operators, so interpret with care).

5.2 ECG / QTc — the reason for this visit

Date	HR	PR	QRS	Raw QT	QTc reported (Bazett)	QTcF (Fridericia)*	Other
17 Mar 2026	—	—	—	—	435 ms	—	sinus, <i>normal QTc on the same drug class</i>
06 Jul 2026	84	—	—	440	520 ms	492 ms	RVH, incomplete RBBB, abnormal Q III
07 Jul 2026	90	—	—	377	463 ms	432 ms	RVH, T-inversion V3–V4
08 Jul 2026	94	—	—	422	529 ms	490 ms	RVH, incomplete RBBB
13 Jul 2026	86	166	86	410	491 ms	462 ms	RVH, poor R progression, low voltage chest
17 Jul 2026	87	143	94	398	481 ms	450 ms	low voltage limb leads

* QTcF calculated by the family from the machine's own printed QT and HR, for discussion only — not a machine output. Recalculating Bazett from those same printed values reproduces the machine's QTc to within 1–2 ms on all five traces, which confirms **every July trace is Bazett-corrected**.

Two observations to put to you: 1. **The raw QT is 377–440 ms throughout — not extreme.** The alarming QTc numbers are substantially an artefact of **Bazett correction at heart rates of 84–94 bpm**, where Bazett systematically over-corrects. On Fridericia the same tracings read **492 → 432 → 490 → 462 → 450 ms**. The direction of travel is favourable on either correction, and the two most recent traces (13 and 17 Jul) are the lowest of the series. 2. Her QTc was **435 ms in March** on the *same* drug class — when her heart rate was slower (avg 75 vs 88 bpm) and her potassium was being actively driven above 4.

5.3 Cardiac biomarkers

Date	NT-proBNP (ref 0–125)	hs-Troponin I (ref ≤15.6 pg/mL)
11 Feb 2026 (outside)	1738	negative
20 Feb 2026	1761	—
08 Mar 2026	—	3.2 (normal)
24 Mar 2026 (PE)	4089	—
25 Mar 2026	—	11.90 (normal)
15 Jun 2026	97.2 (normal)	—
05 Jul 2026	151.4 (mildly high)	—

5.4 Electrolytes & renal — directly relevant to QT and to drug choice

Date	K ⁺	Na ⁺	Creatinine
26 Feb 2026	3.16 (low)	141.1	0.67
13 Mar 2026	3.56	140.2	0.88
16 Mar 2026	3.78	—	—
27 Mar 2026 (AKI)	3.52	135.9	1.74
03 Apr 2026	2.87 (low)	149.8 (high)	0.71
13 Apr 2026	3.57	137.2	0.63
15 Jun 2026	4.7	139	0.93
05 Jul 2026	3.57	136	1.08
07 Jul 2026	3.89	135	1.20

Three flags: - **Potassium is running 3.5–3.9**, i.e. below the **K⁺ > 4** target the RFH team explicitly set for her in March when she was ectopic. She received **IV KCl ×2** and **IV MgSO₄ ×1** on **06 Jul**, and was on a **Lasix infusion** during that admission. - **No serum magnesium level has been measured since 08 Mar 2026** (2.21 mg/dL, normal) — though magnesium was *given* IV on 06 Jul. The same team had set a **Mg²⁺ > 2** target. **She is now on Dytor (torsemide) 20 mg BD plus a tapering prednisolone course — two drugs that actively deplete potassium, and in torsemide's case magnesium too — with a prolonged QT and no magnesium monitoring.** - **Creatinine is climbing steadily:** 0.71 (May) → 0.93 (Jun) → 1.08 (05 Jul) → **1.20 (07 Jul)**. Relevant to apixaban dosing, to diuretic titration, and if any renally-cleared antiarrhythmic (e.g. sotalol) is being contemplated.

5.5 Holter comparison

	14–17 Mar 2026 (RFH)	10–13 Jul 2026 (Surat)
Rhythm	Sinus	Sinus
HR min / avg / max	52 / 75 / 106	78 / 88 / 115
PVCs	1811 (<1 %)	512 (0.1 %)
Couplets	1	5
VT runs	0	0
Bigeminy episodes	60	few
SVE / AF / SVT / AV block / pauses	0 / 0 / 0 / 0 / 0	0 / 0 / 0 / 0 / 0
Symptom correlation	—	1 event (11 Jul) — occurred in NSR

6. Medications

6.1 Cardiac regimen at RFH discharge (08 Apr 2026), per Dr. Talha Meeran — reconfirmed unchanged on 13 Apr

Drug	Dose
Clexane (enoxaparin) 0.6 mL SC	1-0-1 × 14 days, then
Apixaban 5 mg	1-0-1 (continuing)
Cordarone (amiodarone) 200 mg	1-0-1
Mexohar (mexiletine) 150 mg	1-1-1
Aldactone (spironolactone) 50 mg	0-1-0
Tadalafil 20 mg	0-0-1 (for PAH)

Also at discharge: Pan 40 BD, Emeset 8 TDS, Foracort + Duolin nebs, short courses of Effcorlin and Augmentin.

6.2 CURRENT regimen — confirmed from the Maitreya discharge summary of 17 Jul 2026

Cardiac / cardiovascular:

Drug (as printed)	Generic	Dose	Change vs April
Tab Apixalife 5 mg	Apixaban	1-0-1 (BD)	unchanged
Tab Cordarone 100 mg	Amiodarone	1-0-1 (BD)	halved from 200 mg BD
Cap Mexohar 150 mg	Mexiletine	1-0-1 (BD)	reduced from TDS
Tab Aldactone 50 mg	Spironolactone	0-1-0 (OD)	unchanged
Tab Dytor 20 mg	Torsemide	1-1-0 (BD)	NEW — loop diuretic
Tab Tadalafil 20 mg	Tadalafil	0-0-1 (HS)	unchanged (for PAH)

🚫 There is no beta-blocker on this list.

Other drugs that bear on the QT question:

Drug	Why it matters here
Tab Omnacortil (prednisolone) — 40 mg × 5 d → 20 → 10 → 5 → stop	Steroid; wastes potassium
Tab Dytor (torsemide) 20 mg BD	Loop diuretic; wastes potassium AND magnesium
Cap Perinorm CD (metoclopramide) SOS	Itself a QT-prolonging antiemetic
Neb Foracort BD, Tab Doxolin M, Tab DMR, Syp Codistar	Respiratory — she remains bronchospastic
Tab Doxy 100 mg BD, Inj Frastim 300 mcg (14-Jul), Tab Endace 160 mg TDS	Antibiotic / post-chemo G-CSF / appetite
Tab Voqozon 20 mg (vonoprazan), Caltrol D3, Numylin D, Calcore, Threptin, Purple Pro	Acid suppression, vitamins, nutrition

Condition at that discharge (17 Jul): haemodynamically stable, **BP 130/80, PR 83, SpO₂ 93 %** on 1 L/min O₂ support. **Follow-up given:** Dr. Sanjay Agrawal in 5 days; Dr. Kaushal Patel (oncology) in 7 days.

6.3 ★ The dose reduction is recent, deliberate — and appears to be working

Per the family, the **amiodarone and mexiletine doses were cut about 13 July 2026** by the Maitreya team, with Dr. Talha Meeran's **approval**, specifically in response to the prolonged QT.

The ECGs since then support that decision:

	08 Jul	13 Jul	17 Jul
	<i>before/at reduction</i>		
QTc (Bazett)	529 ms	491 ms	481 ms
QTcF (Fridericia)	490 ms	462 ms	450 ms

The QT is already falling on the reduced doses. This matters for today's decision: the question may not be *whether* to act on the QT, but whether an intervention already made ~1 week ago should be given more time before layering another change on top of it.

6.3 ⚠️ A beta-blocker has already been tried — and dropped

Concor (bisoprolol) was started on 06 Mar 2026 alongside amiodarone when the PVCs/NSVT appeared, and it appears in the ward notes through **12 Mar and 20 Mar**. It is **absent from the 08 Apr discharge list and from the 13 Apr daycare list**.

The records do not state why it was stopped. This seems worth establishing before a beta-blocker is re-introduced. Two plausible reasons are visible in the record and both are relevant: - **Recurrent bronchospasm** — documented acute bronchospasm 09 Mar and 20 Mar, and she remained on Foracort/Duolin nebulisers at discharge. - **Severe PAH with a pressure-loaded RV** — beta-blockade is conventionally approached with caution here.

7. Other active problems that constrain cardiac management

- **Bilateral vocal cord palsy**, status post **left vocal cord lateralisation (26 Feb 2026)**. Ongoing aspiration risk; on a modified/pureed diet with safe-swallow precautions. FEES (18 Mar) — improved approximation, no overt aspiration.
- **Recurrent bronchospasm** on inhaled steroid + bronchodilator; **viral LRTI treated 05–08 Jul**, currently on doxycycline and a tapering prednisolone course.
- **Malignant pleural effusions, currently re-accumulating** (HRCT 05 Jul), with a **volume-overload component** (diagnosed 05 Jul, treated with Lasix infusion, now on oral torsemide).
- **Oxygen-dependent again** — off oxygen entirely on 08 Jul, but discharged 17 Jul on 1 L/min at SpO₂ 93 %.
- **Metastatic Ca endometrium in partial response** (PET-CT 22 Jun, –47.1 %), on chemo-immunotherapy — last dose 15 Jun 2026. Oncology now Dr. Kaushal B. Patel, Surat.
- **Prior severe thrombocytopenia and major bleeding** in late March (platelets to 51, haemoptysis, Hb to 7.3) while anticoagulated — relevant to any future antithrombotic decision.
- **Anaemia largely resolved** (Hb 11.4 on 05 Jul); no iron deficiency.

8. Documentation gaps — things the records genuinely do not contain

1. **No thrombectomy procedure note** — no date, no operator, no technical detail. Confirmed only indirectly (§4.2).
2. **No Maitreya discharge summary or prescription** — **RESOLVED**. Both July discharge summaries (08 Jul and 17 Jul) are now in the file, and §6.2 is taken directly from the 17 Jul prescription.
3. **No right heart catheterisation** — PAH severity and classification rest entirely on echo estimates and the CTPA/HRCT appearance.
4. **No coronary angiogram, ever**. It was considered on 08 Mar 2026 and deferred. Her Jul echo explicitly notes that a normal echo does not exclude CAD, and she has documented peripheral atheroma on the May leg Doppler. She is now presenting with **chest pain**.
5. **No serum magnesium level since 08 Mar 2026** — despite a prolonged QT, IV magnesium replacement on 06 Jul, and current torsemide + prednisolone.
6. **No thyroid function tests anywhere in the record** — she has been on amiodarone since 06 Mar 2026, i.e. **about 4.5 months**. TSH/T4 and LFTs are the standard monitoring for that.
7. The **17 Jul ECG carries no patient identifiers** (§4.4).

9. Questions the family would like to put to Dr. Purushottam

These are framed as questions, not opinions — they are the points where the record seems to bear on the decision.

1. **Which of the two drugs is actually prolonging her QT?** Amiodarone is a class III agent whose defining effect is QT prolongation. Mexiletine is class IB and, if anything, tends to **shorten** the QT. Given that, is stopping mexiletine the right lever for a QT problem — or should the amiodarone be the focus?

2. **How much of the “prolonged QTc” is a Bazett artefact?** Her raw QT is 377–440 ms at heart rates of 84–94 bpm, and all five July traces are Bazett-corrected. On Fridericia the last two read 462 and 450 ms. Is that worth confirming before the regimen is changed?
3. **Should potassium and magnesium be optimised first — and is the diuretic load part of the problem?** Her K^+ is 3.57–3.89 against the $K^+ > 4 / Mg^{2+} > 2$ target her Mumbai team set, magnesium has not been *measured* in over four months, and she is now on **torsemide 20 mg BD plus a tapering prednisolone course**, both of which deplete potassium. Is the QT partly a treatable electrolyte problem rather than purely a drug-class problem?
4. **Does the clean 72-hour Holter change the risk calculus?** No VT, no pauses, 0.1 % ectopic burden, and her symptomatic episode occurred in sinus rhythm. Does she still need two antiarrhythmics at all — or is de-escalation of one of them reasonable?
5. **Why was Concor stopped in March, and does that reason still apply?** Given her bronchospasm and pressure-loaded RV, is a cardioselective agent at a low dose the safer option if a beta-blocker is added?
6. **Is amiodarone monitoring due?** ~4.5 months of therapy without TFTs or interval LFTs.
7. **Given ongoing chest pain and a never-excluded coronary question, is any ischaemia evaluation now indicated** — and is it feasible/safe given her renal trend (creatinine 1.20) and oncological status?
8. **Are her current symptoms cardiac at all?** The Holter, the normal troponins, the near-normal NT-proBNP and the recovered RV function all point away from the heart; the increasing pleural effusions on the 05 Jul HRCT point towards the chest. Should the effusions be addressed (drainage?) as the primary symptom driver?
9. **Anticoagulation:** is apixaban still indicated, at what dose, and for how long, given no residual pulmonary thrombus on the 22 Jun PET-CT but an active malignancy and a prior major bleed?

10. Source documents — the complete list used to build this note

All files are in this records archive. Original hospital PDFs are in `report_pdf/` ; the plain-text extractions are in `report_md/` .

Summaries and procedure records

Date	Document	File
08 Apr 2026	Inpatient Discharge Summary, RFH Mumbai (20 Feb – 8 Apr) — the single most important historical document	<code>report_pdf/2026-04-08_Summary-Discharge_Discharge-Summary.pdf</code>
13 Apr 2026	Daycare Discharge Summary (chemotherapy) — reconfirms the April cardiac regimen	<code>report_pdf/2026-04-13_Summary-Daycare_Discharge-Summary-Chemotherapy.pdf</code>
08 Jul 2026	Discharge Summary, Maitreya Surat (05–08 Jul) — viral LRTI, volume overload, SpO ₂ 84 % on air; “Prolonged QT” a listed diagnosis	<code>report_pdf/2026-07-08_Summary-Discharge_Discharge-Summary-Maitreya-Surat.pdf</code>
17 Jul 2026	Discharge Summary, Maitreya Surat (13–17 Jul) — 🌟 the source of the confirmed current medication list in §6.2	<code>report_pdf/2026-07-17_Summary-Discharge_Discharge-Summary-Maitreya-Surat.pdf</code>

Echocardiography

Date	Document	File
20 Feb 2026	Echo Portable — RVSP 83, LVEF 55 %	<code>report_pdf/2026-02-20_ECHO_Echocardiography-Portable.pdf</code>
26 Feb 2026	Echo Portable — RVSP 72, LVEF 60 %	<code>report_pdf/2026-02-26_ECHO_Echocardiography-Portable.pdf</code>
24 Mar 2026	Echo — saddle thrombus, RVSP 91, RV dysfunction FAC 20 %	<code>report_pdf/2026-03-24_ECHO_Echocardiography.pdf</code>
06 Jul 2026	2D Echo Colour Doppler, Maitreya — RVSP 70, LVEF 50 %	<code>report_pdf/2026-07-06_ECHO_2D-Echo-Colour-Doppler-Surat.pdf</code>

ECG and Holter

Date	Document	File
14–17 Mar 2026	72-h Holter, RFH — 1811 PVCs, 60 bigeminy runs, no VT	report_pdf/ 2026-03-14_Holter_Holter-ECG-Report.pdf
14 Mar 2026	ECG Report, RFH — <i>blank; contains no tracing or interpretation</i>	report_pdf/2026-03-14_ECG_ECG-Report.pdf
06 Jul 2026	ECG, Maitreya — QTc 520 ms	report_pdf/2026-07-06_ECG_ECG-Surat.pdf
07 Jul 2026	ECG, Maitreya — QTc 463 ms	report_pdf/2026-07-07_ECG_ECG-Surat.pdf
08 Jul 2026	ECG, Maitreya — QTc 529 ms	report_pdf/2026-07-08_ECG_ECG-Surat.pdf
10–13 Jul 2026	72-h Holter, Surat — sinus, no VT, symptom in NSR	report_pdf/ 2026-07-13_Holter_Holter-ECG-Report-Surat.pdf
13 Jul 2026	ECG, Maitreya — QTc 491 ms (<i>newly filed from family photo</i>)	report_pdf/2026-07-13_ECG_ECG-Maitreya-Surat.pdf
17 Jul 2026	ECG, Maitreya — QTcBz 481 ms (<i>newly filed from family photo</i>)	report_pdf/2026-07-17_ECG_ECG-Maitreya-Surat.pdf

Imaging

Date	Document	File
24 Mar 2026	CT Pulmonary Angiography — saddle PE with RV strain	report_pdf/2026-03-24_CT_CT-Angiography-Pulmonary.pdf
01 May 2026	Right lower limb arterial & venous Doppler — no DVT	report_pdf/2026-05-01_USG_Right-Lower-Limb-Arterial-Venous-Doppler.pdf
22 Jun 2026	WB FDG PET-CT — RECIST PR –47.1 %; no residual pulmonary thrombus	report_pdf/2026-06-22_CT_WB-FDG-PET-CT-scan.pdf
05 Jul 2026	HRCT Chest — effusions increased; PA 3.4 cm	report_pdf/2026-07-05_CT_HRCT-Chest-Plain-Surat.pdf

Laboratory

Date	Document	File
20 Feb 2026	NT-proBNP 1761	report_pdf/2026-02-20_Lab-Serol_NT-pro-B-type-Natriuretic-Peptide-NT-pro-BNP.pdf
08 Mar 2026	hs-Troponin I 3.2 (normal)	report_pdf/2026-03-08_Lab-Biochem_Troponin-I-hs-Less-than.pdf
08 Mar 2026	Magnesium 2.21 — the last magnesium on record	report_pdf/2026-03-08_Lab-Serol_Magnesium.pdf
24 Mar 2026	NT-proBNP 4089 (peak)	report_pdf/2026-03-24_Lab-Serol_NT-pro-B-type-Natriuretic-Peptide-NT-pro-BNP.pdf
25 Mar 2026	hs-Troponin I 11.90 (normal, at the time of the PE)	report_pdf/2026-03-25_Lab-Biochem_Troponin-I-hs.pdf
31 Mar 2026	Blood culture — “post thrombectomy”, left IJV, no growth	report_pdf/2026-03-31_Lab-Micro_Blood-Culture-Set.pdf
15 Jun 2026	Combined panel — NT-proBNP 97.2 (normal) , creatinine 0.93	report_pdf/2026-06-15_Lab-Surat_Combined-Panel.pdf
05 Jul 2026	Admission panel — NT-proBNP 151.4, CRP 9.0, creatinine 1.08, K 3.57	report_pdf/2026-07-05_Lab-Surat_Combined-Panel.pdf
06 Jul 2026	Iron profile + ferritin — no iron deficiency	report_pdf/2026-07-06_Lab-Surat_Iron-Profile-Ferritin.pdf
06 Jul 2026	Urine routine — normal	report_pdf/2026-07-06_Lab-Surat_Urine-Routine.pdf
07 Jul 2026	Creatinine 1.20, Na 135, K 3.89	report_pdf/2026-07-07_Lab-Surat_Creatinine-Electrolytes.pdf

Consolidated references

Document	File
All serial lab values (Hb, electrolytes, creatinine, CRP, NT-proBNP, coagulation) in dated tables	TRENDS.md
Date-grouped index of every report in the archive	MASTER_INDEX.md
Raw original Surat PDFs, incl. the 16-page July Maitreya bundle	surat_reports_and_other_details/
Maitreya pharmacy invoices — source for the IV KCl / MgSO ₄ / Lasix given 05–08 Jul (<i>held locally, not in the published archive</i>)	admin_and_billing/2026-07_Medicine-Invoice_Maitreya.pdf
Maitreya IPD invoice and insurance Claim Form Part B — confirms emergency admission (<i>held locally</i>)	admin_and_billing/

Compiled 20 July 2026 from the primary records listed above. Figures are quoted from source documents; inferences, gaps and inconsistencies are marked ⚠️. Current medication doses in §6.2 are the family’s account and are not supported by a document in the file.